

5

TIME AND DISTANCE

5.1. Speed

We have the relation between speed, time and distance as follows:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

So the distance covered in unit time is called speed.

This forms the basis for Time and Distance. It can be re-written as Distance = Speed X Time or

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}.$$

5.1.1. Units of Speed

The units of speed are kmph (km per hour) or m / s.

$$1 \text{ kmph} = \frac{5}{18} \text{ m/s}$$

$$1 \text{ m/s} = \frac{18}{5} \text{ kmph}$$

5.1.2. Average Speed

When the travel comprises of various speeds then the concept of average speed is to be applied.

$$\text{a Average Speed} = \frac{\text{Total distance covered}}{\text{Total time of travel}}$$

Note: In the total time above, the time of rest is not considered.

Example 1: If a car travels along four sides of a square at 100 kmph, 200 kmph, 300 kmph and 400 kmph find its average speed.

Solution: Average Speed = $\frac{\text{Total distance}}{\text{Total time}}$.

Let each side of square be x km. Then the total distance = $4x$ km.

The total time is sum of individual times taken to cover each side.



To cover x km at 100 kmph, time = $\frac{x}{100}$.

For the second side time = $\frac{x}{200}$.

Using this we can write average speed = $\frac{4x}{\left(\frac{x}{100} + \frac{x}{200} + \frac{x}{300} + \frac{x}{400}\right)} = 192 \text{ kmph}$.

Example 2: A man if travels at $\frac{5}{6}$ th of his actual speed takes 10 min more to travel a distance. Find his usual time.

Solution: Let s be the actual speed and t be the actual time of the man.

Now the speed is $\left(\frac{5}{6}\right)s$ and time is $(t+10)$ min. But the distance remains the same.

So distance 1 = distance 2 $\Rightarrow s \times t = \left(\frac{5}{6}\right)s \times (t+10) \Rightarrow t = 50 \text{ min}$.

Example 3: If a person walks at 30 kmph he is 10 min late to his office. If he travels at 40 kmph then he reaches to his office 5 min early. Find the distance to his office.

Solution: Let the distance to his office be d . The difference between the two timings is given as 15 min = $\frac{1}{4}$ hr.

Now if d km are covered at 30 kmph then time = $d/30$. Similarly second time = $d/40$.

So, $\frac{d}{30} - \frac{d}{40} = \frac{1}{4} \Rightarrow d = 30 \text{ km}$.

Note: When two objects move with speeds s_1 and s_2

- In opposite directions their combined speed = $s_1 + s_2$
- In same direction their combined speed = $s_1 \sim s_2$.

Example 4: Two people start moving from the same point at the same time at 30 kmph and 40 kmph in opposite directions. Find the distance between them after 3 hrs.

Solution: Speed = $30 + 40 = 70 \text{ kmph}$ (since in opposite directions)

Time = 3 hrs

So distance = speed $\times$ time = $70 \times 3 = 210 \text{ km}$.

Example 5: A starts from X to Y at 6 am at 40 kmph and at the same time B starts from Y to X at 50 kmph. When will they meet if X and Y are 360 km apart?

Solution: Distance = 360 km, Speed = $40 + 50 = 90 \text{ kmph}$.

Time = $\frac{\text{distance}}{\text{speed}} = \frac{360}{90} = 4 \text{ hrs from 6 am} \Rightarrow 10 \text{ am}$.

